

Complete A-Z Website, CRM, CMS & Media Operations Specification

Business logic + UI/UX + motion + admin uploads + media processing + CRM + technical architecture + deployment

Build a premium videography-agency platform that sells the service, manages the content, captures leads, runs the sales pipeline, and lets a non-technical owner upload new videos, photos, thumbnails, colour-grade assets and complete projects without touching code.

Placeholder	Replace during discovery
[AGENCY NAME]	Fix Frame / final approved brand name
[TAGLINE]	Final brand proposition
[DOMAIN]	www.example.com
[LOCATION]	Primary city / service area
[PHONE]	+91 XXXXX XXXXX
[EMAIL]	hello@example.com
[INSTAGRAM]	@handle
[RESPONSE TIME]	e.g. within 24 business hours
[BUDGET RANGES]	Final lead qualification brackets
[BRAND FONT]	Final licensed/approved display font

How to use this document

- Anything inside square brackets is intentionally editable placeholder content.
- Public website presentation is controlled by code/templates. CMS users manage content, not layout or animation settings.
- Every upload workflow is designed for a non-technical operator.
- Business rules define what the company needs; technical rules define how developers enforce it.
- Phase 1 can launch lean. Advanced features are clearly marked as later/optional.

Document map

Part	Area	What it covers
0	Product premise	Scope, users, success metrics, phased rollout
A	Business model	Services, sales funnel, operating model
B	Design system	Theme, type, grid, motion, wave language
C	Public website	Every customer-facing page and form
D	Lead generation	Qualification, validation, pipeline creation
E	CRM	Dashboard, leads, clients, projects, calendar, tasks, finance
F	CMS & media studio	Videos, photos, thumbnails, colour grade, projects, reels
G	Workflow engine	Project lifecycle from enquiry to delivery
H	Technical architecture	Next.js, ASP.NET Core, DB, storage, queues, CDN
I	Data & API contracts	Entities, endpoints, permissions, events
J	Security & compliance	Auth, uploads, privacy, backups, audit
K	Performance / SEO / analytics	CWV, metadata, conversion tracking
L	Deployment / cost	Environments, hosting, domain, operations
M	Build phases & QA	Step-by-step implementation, tests, handover

Product Premise

One connected system: premium public experience + lead engine + content/media management + CRM/project operations.

0.1 Product outcomes

Outcome	Business value	Product response
Premium perception	Higher-quality enquiries and stronger pricing power	Editorial motion-first website using real work
Lead quality	Less time wasted on vague enquiries	Structured project-brief form and qualification
Owner independence	No developer needed for routine content	Guided CMS upload and publish flows
Sales visibility	Know exactly where every enquiry sits	CRM pipeline with statuses, notes and tasks
Production control	Reduce missed shoot/edit/delivery tasks	Project lifecycle, milestones, files and calendar
Content freshness	New work can be published quickly	Project/reel/media studio with processing

0.2 User roles

Role	Primary responsibilities	Typical permissions
Visitor	Explore work and enquire	Public read + submit form
Owner/Admin	Full business control	All CMS/CRM/settings/users
Content Editor	Publish media and projects	CMS/media, no finance/users
Sales	Manage leads/clients/quotes	CRM leads, clients, tasks
Production	Manage active projects	Projects, calendar, files, tasks
Finance	Invoices/payments	Finance-only modules

0.3 Release strategy

Release	Must include	Can wait
MVP	Public site, lead form, CRM leads, projects, video/photo upload, thumbnails, basic CMS	Quotation PDFs, payments, client portal
V1.1	Advanced project workflow, calendar/tasks, testimonials/packages	Automations, advanced reports
V2	Quotations, invoice/payment tracking, client review portal	AI-assisted tools, complex integrations

Business Model & Operating Logic

Translate the agency's service delivery into clear website conversion and internal operational workflows.

A1. Service model

Service family	Examples	Lead qualification questions
Videography	Wedding, event, commercial, corporate	Date, location, duration, crew, deliverables, budget
Photography	Event, portrait, product, campaign	Date, location, number of looks/products, delivery
Post-production	Editing, colour, sound, motion	Footage size, duration, format, deadline, references
Drone	Aerial shoot / add-on	Location, permission constraints, shoot window
Social / Reels	Short-form packages	Volume/month, platforms, shoot/edit split

A2. Conversion funnel

DISCOVER → IMPRESS → SHOW RELEVANT WORK → BUILD TRUST → QUALIFY → CREATE LEAD → CONTACT → PROPOSAL → WON → PROJECT → DELIVERY

- 01 Visitor lands from search, social, referral or direct link.
- 02 Hero and selected work establish quality immediately.
- 03 Visitor explores relevant project/category/service.
- 04 Proof points: real BTS, testimonials, clients, process.
- 05 Visitor chooses Start a Project / Get a Quote.
- 06 Project brief captures enough information for sales triage.
- 07 System creates Lead first, then sends notifications.
- 08 Sales contacts lead and moves status through pipeline.
- 09 Won lead converts to Client + Project.
- 10 Production workflow continues until final delivery and payment.

A3. Business rules that developers must preserve

- A public enquiry is never lost because email notification failed; database persistence happens first.
- A lead can exist without a project. A project is created only when intentionally converted.
- Published portfolio work must have required poster/cover media and approved rights.
- Raw camera masters are not required in the website CMS; publish optimized delivery assets.
- Colour-grade functionality in MVP is asset management and comparison, not a replacement for professional grading software.
- Client-facing site never exposes internal notes, budget discussion history or file-storage keys.

Design System, Motion & Brand Language

Create a recognisable human-directed visual language instead of a generic AI/SaaS layout.

B1. Core visual direction

System	Rule
Canvas	Near black / off-black; avoid random gradients
Accent	Brand red used selectively for action and editorial emphasis
Wave language	Thin waveform-like paths, progress lines, timeline curves; not decorative blobs everywhere
Type	One strong display family + one neutral sans; final fonts [TBD]
Cards	Use only for functional CRM grouping; public site favors editorial composition
Media	Real work first; no fake AI portfolio
Radius	Mostly 0-8px; avoid universal pill/rounded-card look

B2. Token proposal

Token	Value	Usage
--canvas	#080808	Primary dark background
--surface	#101010	Functional panel
--surface-2	#161616	Raised panel
--line	#2A2A2A	Hairline
--ink	#F3F1ED	Primary text
--ink-soft	#B8B3AD	Secondary text
--muted	#77736F	Metadata
--accent	#D61F2C	Primary action
--success	#2F8F62	Won/success
--info	#3A77B9	Contacted/info
--warn	#B96B22	Proposal/pending
--discussion	#7758A6	Discussion/negotiation

B3. Motion rules

Motion	Duration	Where
Fast	160-200ms	Hover, opacity, borders
Standard	300-450ms	Menus, panels, filters
Editorial	700-1000ms	Text/media reveals
Cinematic	1000-1600ms	Hero/project transitions
Wave progress	600-1200ms	CRM charts/timeline/status transitions

- Wave motion communicates flow/progress; it should respond to real data where possible.
- Status colors are never shown only as text color; pair with label/icon/shape.
- prefers-reduced-motion removes pinning, parallax and animated waveform travel.

Public Website Screens

A premium landing experience that sells the service before the CRM ever becomes relevant.

Home / Cinematic Landing

Establish premium positioning in seconds, surface selected work, demonstrate editing craft and drive qualified project enquiries.

Area	Specification
Primary user	Prospective client
Primary goal	Perceive quality and start enquiry
Primary CTA	[START A PROJECT]
Secondary CTA	[VIEW WORK]
Data owner	CMS
Permission	Public

Screen structure & content

Hero. [AGENCY NAME] / [TAGLINE]. Full-bleed real showreel or selected project footage. CTA: [VIEW WORK] and [START A PROJECT].

Selected Stories. 3-6 featured projects with asymmetrical editorial layouts, project metadata and hover/tap preview.

Editing Signature. RAW → EDIT → GRADE → SOUND → FINAL. Use real paired assets.

Services. Oversized typographic service chapters, not six identical cards.

Process. Discover → Plan → Create → Edit → Deliver.

Proof. Real client logos, testimonials, BTS, awards/stats only when verifiable.

Final CTA. [HAVE A STORY IN MIND?] Start a Project.

User interactions

- 01** Load hero poster immediately, then play optimized muted video when ready.
- 02** View Work scrolls/navigates to Work.
- 03** Start a Project opens dedicated multi-step brief.
- 04** Project item hover shows muted short preview after intent delay.
- 05** Editing section lets user compare raw/final manually even when motion is disabled.

Business logic

- Only published projects marked Featured may appear on home.
- Hero media requires poster; fallback poster must preserve composition.
- Testimonials/brand logos require approval before public display.
- Do not auto-play audio.

Technical implementation

- Next.js page uses server-rendered published CMS data.

- Critical media preloading limited to hero poster and one hero video variant.
- GSAP timelines encapsulated in reusable components; CMS contains no animation fields.
- Analytics events: hero_cta, work_open, showreel_play, lead_form_start.

States

State	Required behavior
Loading	Poster and stable layout appear; video can arrive later.
Empty	If no featured projects, show curated fallback project set configured by admin.
Error	Media failure falls back to poster; page remains usable.
Success	All CTAs route correctly and content is readable before animation.

Responsive & motion

- Desktop can use pinned/editorial transitions; mobile uses vertical narrative.
- Wave motif may appear as a thin transition/progress line between chapters, not as a background decoration.
- Reduced motion uses opacity-only transitions.

Acceptance criteria

- No placeholder project remains in production.
- Mobile hero loads within agreed performance budget.
- All CTA paths tested.
- Keyboard and reduced-motion behavior verified.

Work / Portfolio

Help visitors find relevant proof quickly without turning the site into a generic gallery grid.

Area	Specification
Primary user	Prospective client
Primary goal	Find relevant work
Primary CTA	[VIEW PROJECT]
Secondary CTA	[START A PROJECT]
Data owner	Projects CMS
Permission	Public

Screen structure & content

Intro. [SELECTED WORK] + short positioning line.

Filters. All / Weddings / Commercial / Events / Editing / Photography / [CUSTOM].

Project stream. Editorial list/masonry with title, category, year, optional client.

CTA. [START A SIMILAR PROJECT].

User interactions

- 01** Filter work.
- 02** Open project detail.
- 03** Play preview where available.
- 04** Start enquiry prefilled with project/category context.

Business logic

- Only Published projects are returned.
- Category filters do not change canonical project URLs.
- Featured order can override date order.

Technical implementation

- GET /api/public/projects?category=&cursor=.
- Use pagination/cursor loading.
- Preview video only loads on interaction/near viewport.

States

State	Required behavior
Loading	Skeleton/media placeholders preserve proportions.

Empty	Friendly category-empty message + All Work action.
Error	Retry and keep current filter.
Success	URL/filter state can be shareable if implemented.

Responsive & motion

- Mobile uses single/dual column depending asset ratio.
- Hover-only metadata becomes persistent.
- Filter bar becomes horizontal scroll or compact select.

Acceptance criteria

- Filter results accurate.
- No unpublished content leaks.
- Media lazy-loading verified.
- Project links preserve SEO routes.

Project Case Study

Convert beautiful work into evidence of capability and a relevant sales CTA.

Area	Specification
Primary user	Prospective client
Primary goal	Evaluate specific capability
Primary CTA	[START A SIMILAR PROJECT]
Secondary CTA	[NEXT STORY]
Data owner	Project CMS
Permission	Public

Screen structure & content

Hero. Project title, category, year, cover video/image.

Metadata. Client, location, services, duration/format where relevant.

Story. Brief, creative approach, outcome.

Media blocks. Video, gallery, BTS, before/after, credits.

Testimonial. Optional approved quote.

Next story. Editorial next-project navigation.

CTA. [WANT SOMETHING LIKE THIS?].

User interactions

- 01** Play video.
- 02** Open gallery.
- 03** Compare raw/final if configured.
- 04** Open next project.
- 05** Start enquiry with sourceProjectId.

Business logic

- Project must be Published.
- Project blocks follow approved block types only.
- Private/internal media never returned by public API.

Technical implementation

- SSR metadata generated from Project SEO fields.
- Video/gallery blocks loaded progressively.
- Structured VideoObject markup when applicable.

States

State		Required behavior
Loading		Hero poster first.
Empty		Optional sections simply omitted; required hero/title enforced in CMS.
Error		404 if unpublished/not found.
Success		Related CTA contains project context.

Responsive & motion

- Editorial composition simplifies on mobile.
- Video controls remain accessible.
- No scroll trapping.

Acceptance criteria

- SEO metadata valid.
- All public media authorized.
- CTA prefill tested.
- 404/unpublished behavior tested.

Services

Explain what the studio sells and make it easy for a prospect to start the right enquiry.

Area	Specification
Primary user	Prospective client
Primary goal	Understand offer
Primary CTA	[GET A QUOTE]
Secondary CTA	[VIEW WORK]
Data owner	Services CMS
Permission	Public

Screen structure & content

Service chapters. Videography, Photography, Editing, Drone, Social, Commercial.

Deliverables. [PLACEHOLDER BULLETS] per service.

Featured work. Relevant projects.

Process. How engagement works.

CTA. [GET A QUOTE].

User interactions

- 01** Open service detail/expand chapter.
- 02** Open relevant project.
- 03** Start project brief with service preselected.

Business logic

- Only active services show.
- Packages may be hidden independently.
- CTA preselects service.

Technical implementation

- Services endpoint returns active ordered content.
- Related work query by service/category.

States

State	Required behavior
Loading	Text first, media progressively.
Empty	Do not show empty service chapters.

Error

Fallback to contact CTA.

Success

Correct service passed to enquiry.

Responsive & motion

- Desktop media can react to service hover/scroll.
- Mobile uses stacked sections.
- Reduced motion removes media parallax.

Acceptance criteria

- Every service has clear CTA.
- No duplicate generic copy.
- Mobile readable.

Editing / Post-production

Demonstrate post-production as a premium service using real before/after material rather than generic claims.

Area	Specification
Primary user	Editing prospect
Primary goal	Trust post-production quality
Primary CTA	[GET AN EDITING QUOTE]
Secondary CTA	[VIEW EDITING WORK]
Data owner	Editing CMS
Permission	Public

Screen structure & content

Hero. [THE STORY IS MADE IN THE EDIT.]

Raw-to-final sequence. Raw / edit / grade / sound / final.

Before/after pairs. Real approved frame/video comparisons.

Capabilities. Offline edit, colour, sound, motion, reels, long-form.

Workflow. Receive footage → Brief → First cut → Review → Master.

CTA. [SEND US YOUR FOOTAGE BRIEF].

User interactions

- 01** Compare raw/final.
- 02** Play editing reel.
- 03** Open editing project.
- 04** Start editing enquiry.

Business logic

- Raw/final pairs must be explicitly paired in CMS.
- MVP stores graded final and raw preview; no destructive grade applied by website.
- Reference to LUT/profile is metadata only unless optional processor enabled.

Technical implementation

- BeforeAfterPair entity or project media relationship.
- Video variants optimized independently.
- Optional later: LUT preview service using ffmpeg/processing worker.

States

State	Required behavior
Loading	Show posters until media ready.
Empty	Hide comparison module if no approved pair.
Error	Fall back to final media only.
Success	Pair sync and labels accurate.

Responsive & motion

- Mobile uses slider/toggle, not complex pinned sequence.
- Reduced motion keeps manual comparison.

Acceptance criteria

- Raw/final labels accurate.
- No internal master exposed.
- Enquiry prefilled with Editing.

About / Studio

Build trust through real people, BTS and operational credibility.

Area	Specification
Primary user	Prospective client
Primary goal	Trust team
Primary CTA	[START A PROJECT]
Secondary CTA	[CONTACT]
Data owner	Site CMS
Permission	Public

Screen structure & content

Story. [WHY THE STUDIO EXISTS].

Team. [NAME / ROLE / BIO].

BTS. Real production stills/film.

Values. [STORY / CRAFT / RELIABILITY / COLLABORATION].

Stats. Only verifiable metrics.

CTA. [START A PROJECT].

User interactions

01 Explore BTS.

02 Open contact/enquiry.

Business logic

- Team/status entries manually managed.
- Client logos/stats approved before publication.

Technical implementation

- About content in SitePage or structured sections.
- BTS media references MediaAsset.

States

State	Required behavior
Loading	Text first.
Empty	Omit optional blocks.

Error

Keep basic contact path.

Success

All content approved.

Responsive & motion

- Single-column mobile narrative.
- Minimal motion.

Acceptance criteria

- No fake stats.
- Team data current.
- Contact path works.

Packages

Provide useful price anchoring while keeping custom-scope flexibility.

Area	Specification
Primary user	Price-aware prospect
Primary goal	Understand starting investment
Primary CTA	[REQUEST QUOTE]
Secondary CTA	[CUSTOM QUOTE]
Data owner	Packages CMS
Permission	Public

Screen structure & content

Package cards/rows. [ESSENTIAL], [SIGNATURE], [PREMIUM], [EDITING-ONLY].

Price. Starting [₹XX,XXX] or [CUSTOM QUOTE].

Inclusions. Deliverables / hours / revisions / travel note.

CTA. [REQUEST THIS PACKAGE] / [CUSTOM QUOTE].

User interactions

01 Select package.

02 Start project brief with package/service preselected.

Business logic

- Prices are display strings in MVP, not invoicing source of truth.
- Inactive packages hidden.
- Travel/tax disclaimer configurable.

Technical implementation

- GET /api/public/packages.
- Store display price and inclusions.
- Pass packageId into lead form.

States

State	Required behavior
Loading	Stable rows/cards.
Empty	Hide packages route or show custom quote only.

Error

Fallback to contact.

Success

Selected package carried into lead.

Responsive & motion

- Stack on mobile.
- Avoid wide comparison tables.

Acceptance criteria

- Owner can change package without deployment.
- Lead receives package context.

Start a Project

Capture a qualified brief while keeping the form easy enough to complete on mobile.

Area	Specification
Primary user	Prospective client
Primary goal	Submit qualified enquiry
Primary CTA	[SEND PROJECT BRIEF]
Secondary CTA	[SAVE? optional later]
Data owner	Lead system
Permission	Public

Screen structure & content

Step 1. Service / project type.

Step 2. Date / location / duration.

Step 3. Budget range.

Step 4. Idea / deliverables / reference links.

Step 5. Name / email / phone / preferred contact.

Consent. Privacy / optional marketing.

Success. Reference number + response-time promise.

User interactions

- 01** Select service.
- 02** Navigate forward/back without losing data.
- 03** Submit.
- 04** Use direct WhatsApp after successful submission if desired.

Business logic

- At least one contact method required.
- Database insert succeeds before notifications.
- Duplicate detection may flag same email/phone + recent timeframe, but must not silently discard.
- Service-specific questions can appear conditionally.
- Budget includes [PREFER NOT TO SAY].

Technical implementation

- Client-side validation mirrors server validation but server is authoritative.
- POST /api/public/leads with idempotency token.
- Rate limit + anti-bot.

- Notification job queued after commit.

States

State	Required behavior
Loading	Submit button shows progress and disables duplicate click.
Empty	Not applicable.
Error	Keep all answers, focus first error, show retry.
Success	LeadId/reference returned; analytics fires only after persisted.

Responsive & motion

- One logical question group per mobile viewport.
- No animation while typing.
- Progress bar can use subtle wave travel.

Acceptance criteria

- No data loss on validation.
- Duplicate submit protected.
- Lead visible in CRM immediately.
- Notification failure does not delete lead.

Lead Generation Logic

Formal qualification, source tracking and CRM creation rules.

D1. Lead form field matrix

Field	Type	Required	Business rule	CRM mapping
Service	select	Yes	One primary service	Lead.ServiceId
Project type	select/text	Yes	Service-dependent options	Lead.ProjectType
Date	date	No	Can be unknown	Lead.ProjectDate
Location	text	Recommended	Free text + optional normalized city later	Lead.Location
Budget	range	No	Include prefer-not-to-say	Lead.BudgetRange
Idea	textarea	Yes	Min/max length	Lead.Brief
Reference URLs	list	No	Allow safe http/https only	LeadReference
Name	text	Yes	Person/company accepted	Lead.Name
Email	email	Conditional	Email or phone required	Lead.Email
Phone	tel	Conditional	Normalize country code where possible	Lead.Phone
Preferred contact	enum	No	WhatsApp/Call/Email	Lead.PreferredContact
Source	system	Yes	Page/project/campaign	Lead.Source

D2. Lead status contract

Status	Meaning	Allowed next	Color
New	Uncontacted valid lead	Contacted / Lost	D61F2C
Contacted	Initial outreach completed	Qualified / Lost	3A77B9
Qualified	Real opportunity	Proposal / Lost	7758A6
Proposal Sent	Commercial sent	Negotiation / Won / Lost	B96B22
Negotiation	Scope/price discussion	Won / Lost	7758A6
Won	Confirmed customer	Convert to project	2F8F62
Lost	Not proceeding	Reopen	666666

CRM Screens & Business Operations

A functional internal system for sales and production, using the wave language for data/progress rather than decoration.

CRM Dashboard

Give owner/manager a fast operational summary of leads, projects, follow-ups and revenue indicators.

Area	Specification
Primary user	Owner / Manager
Primary goal	Know what needs attention
Primary CTA	[VIEW NEW LEADS]
Secondary CTA	[NEW PROJECT]
Data owner	CRM
Permission	Admin/Sales/Production filtered

Screen structure & content

KPIs. New leads, qualified, projects in progress, pending follow-ups, invoices due, [revenue if enabled].

Lead wave. Time-series line/wave showing lead volume and conversion.

Pipeline. Counts by lead status.

Activity. Recent lead/project actions.

Upcoming. Calls, shoots, deadlines.

User interactions

- 01 Change date range.
- 02 Open KPI drill-down.
- 03 Open lead/project/activity.
- 04 Create lead/project if permission allows.

Business logic

- KPIs are calculated from live data and current date range.
- Finance cards hidden from roles without permission.

Technical implementation

- Dashboard aggregation endpoints or optimized queries.
- Cache short-lived aggregates.
- Charts use accessible labels and data table fallback if needed.

States

State	Required behavior
-------	-------------------

Loading	Skeletons per widget.
Empty	Show zero + next action, not broken charts.
Error	Individual widgets can fail independently.
Success	KPI click routes to filtered list.

Responsive & motion

- Wave chart animates only on initial reveal; reduced motion static.
- Mobile stacks KPIs and replaces dense chart with compact summary.

Acceptance criteria

- Counts match source lists.
- Role permissions verified.
- Date-range filters consistent.

Leads List

Manage all enquiries with fast filtering, clear status and minimal visual noise.

Area	Specification
Primary user	Sales / Admin
Primary goal	Prioritize and progress leads
Primary CTA	[NEW LEAD]
Secondary CTA	[FILTERS]
Data owner	CRM
Permission	Sales/Admin

Screen structure & content

Header. Search + New Lead.

Status tabs. All / New / Contacted / Qualified / Proposal / Negotiation / Won / Lost.

Filters. Project type, budget, source, assignee, date.

Table. Lead, project, budget, status, assignee, last contact, created date.

Quick drawer. Optional right-side lead preview.

User interactions

- 01** Search/filter.
- 02** Open lead.
- 03** Change status inline if permitted.
- 04** Assign owner.
- 05** Create manual lead.

Business logic

- Status filter source of truth is Lead.Status.
- Lost leads remain searchable.
- Manual lead must store source=Manual.

Technical implementation

- Server-side pagination/filtering.
- Debounced search.
- Optimistic status updates only with rollback on failure.

States

State	Required behavior
-------	-------------------

Loading	Table skeleton.
Empty	Status-specific empty message.
Error	Keep filters and retry.
Success	Changes reflected in detail/history.

Responsive & motion

- Desktop table; mobile card list.
- Status wave underline/marker can visually flow, but text badge remains explicit.

Acceptance criteria

- Pagination accurate.
- Filters composable.
- No unauthorized bulk actions.

Lead Detail

Provide complete enquiry context, notes, history and conversion actions.

Area	Specification
Primary user	Sales / Admin
Primary goal	Move lead toward close
Primary CTA	[MARK AS CONTACTED / NEXT STATUS]
Secondary CTA	[CONVERT TO PROJECT]
Data owner	CRM
Permission	Sales/Admin

Screen structure & content

Identity. Name/company, contact, source, created time.

Project brief. Service, type, date, location, budget, idea, references.

Tabs. Details / Notes / Tasks / Files / History.

Assignment. Owner/salesperson.

Actions. Contact, status change, convert to project, mark lost.

User interactions

- 01** Call/email/WhatsApp.
- 02** Add note.
- 03** Create task.
- 04** Attach file.
- 05** Change status.
- 06** Convert won lead to project.

Business logic

- Conversion requires Won status or explicit authorized override.
- Conversion is transactional: create/reuse Client + create Project + link Lead.
- Internal notes never emailed automatically.

Technical implementation

- POST /api/admin/leads/{id}/notes.
- POST /api/admin/leads/{id}/status.
- POST /api/admin/leads/{id}/convert.
- Activity log appended on state-changing actions.

States

State	Required behavior
Loading	Section skeleton.
Empty	No notes/tasks/files states with add action.
Error	Failed mutation keeps entered note/task.
Success	History shows action immediately.

Responsive & motion

- Side panel can become full screen on mobile.
- Wave/progress indicator may show pipeline stage at top.

Acceptance criteria

- Conversion idempotent.
- History complete.
- Permissions enforced.

Clients

Maintain customer records after conversion and support repeat business.

Area	Specification
Primary user	Sales/Admin
Primary goal	Manage customer history
Primary CTA	[ADD CLIENT]
Secondary CTA	[NEW PROJECT]
Data owner	CRM
Permission	Sales/Admin

Screen structure & content

List. Client/company, contact, projects, total invoiced [if finance], last project.

Detail. Contacts, projects, notes, files, financial summary [permission].

User interactions

- 01** Search.
- 02** Open client.
- 03** Create manual client.
- 04** Start new project for client.

Business logic

- Client may have multiple contacts/projects.
- Deleting client should be archive, not hard delete when referenced.

Technical implementation

- Client/ClientContact entities.
- Aggregate project counts and finance only when authorized.

States

State	Required behavior
Loading	List skeleton.
Empty	Add first client.
Error	Retry.
Success	Project relationships accurate.

Responsive & motion

- Mobile cards.
- No complex wave decoration needed.

Acceptance criteria

- Repeat project creation works.
- Archived clients hidden by default.

Projects / Pipeline

Track active work from newly won business through final delivery.

Area	Specification
Primary user	Production/Admin
Primary goal	Control active work
Primary CTA	[NEW PROJECT]
Secondary CTA	[VIEW CALENDAR]
Data owner	CRM
Permission	Production/Admin

Screen structure & content

Views. Kanban / List / Calendar.

Stages. Pre-production / Scheduled / Production / Editing / Review / Final Delivery / Completed / On Hold.

Card. Project, client, next milestone, owner, due date, payment flag [optional].

User interactions

- 01** Drag stage where permitted.
- 02** Open project.
- 03** Create project.
- 04** Filter by owner/service/date.

Business logic

- Stage changes logged.
- Certain stage transitions may require milestone checks later, but MVP can warn instead of block.
- Completed projects remain accessible.

Technical implementation

- Project.Stage enum/state machine.
- Drag-drop calls PATCH stage; rollback on failure.
- Kanban query grouped by stage.

States

State	Required behavior
Loading	Column skeletons.
Empty	Empty columns remain valid.

Error	Retry per view.
Success	History updated.

Responsive & motion

- Mobile defaults to list or horizontal staged view.
- Wave header can visualize overall pipeline flow.

Acceptance criteria

- Stage counts accurate.
- Drag preserves order if implemented.
- No data loss.

Project Detail

Central operational record for production, editing, files, tasks and client history.

Area	Specification
Primary user	Production/Admin
Primary goal	Deliver project reliably
Primary CTA	[ADD TASK / UPDATE STAGE]
Secondary CTA	[UPLOAD MEDIA]
Data owner	CRM
Permission	Production/Admin

Screen structure & content

Overview. Client, service, date, location, stage, budget/quote reference.

Timeline. Milestones with wave/progress line.

Tabs. Overview / Tasks / Calendar / Files / Media / Notes / Finance / History.

Milestones. Brief approved, proposal, booking, shoot, first cut, review, final, payment.

User interactions

- 01 Change stage.
- 02 Add milestone/task/note/file.
- 03 Open public portfolio draft if linked.
- 04 Create invoice/quote if enabled.

Business logic

- Project operational media and public portfolio media are related but not identical.
- Completing project does not automatically publish it as portfolio work.
- Project can later be converted into a public portfolio ProjectContent entry.

Technical implementation

- Project entity + Task/Milestone/File relationships.
- Media relationship supports internal/public visibility flag.
- Activity log captures major mutations.

States

State	Required behavior
Loading	Tab skeleton.

Empty	Contextual add actions.
Error	Mutation-specific error.
Success	Progress and stage recalc.

Responsive & motion

- Timeline wave collapses into vertical milestone list on mobile.

Acceptance criteria

- Public/internal visibility separation tested.
- Milestones persist.
- Role restrictions work.

Calendar

Show shoots, meetings, deadlines and task dates in one operational calendar.

Area	Specification
Primary user	Sales/Production
Primary goal	Avoid schedule conflicts
Primary CTA	[NEW EVENT]
Secondary CTA	[TODAY]
Data owner	CRM
Permission	Role-based

Screen structure & content

Views. Month / Week / Day.

Events. Shoot, meeting, edit deadline, delivery, payment due [if enabled].

Filters. Team, project, event type.

User interactions

- 01 Open event.
- 02 Create event.
- 03 Drag/reschedule if permitted.
- 04 Open linked project/lead.

Business logic

- Events linked to project/lead when possible.
- Rescheduling logs history.
- Time zone [TBD] consistent.

Technical implementation

- CalendarEvent entity or integration adapter later.
- Server stores timezone-aware timestamps.

States

State	Required behavior
Loading	Calendar placeholders.
Empty	Normal empty calendar.

Error

Retry without losing selected date.

Success

Linked entity opens correctly.

Responsive & motion

- Mobile defaults agenda/day.
- No drag-only requirement.

Acceptance criteria

- Timezone tests pass.
- Reschedule audit stored.

Tasks

Centralise follow-ups and production work items.

Area	Specification
Primary user	All internal roles
Primary goal	Complete assigned work
Primary CTA	[NEW TASK]
Secondary CTA	[MY TASKS]
Data owner	CRM
Permission	Role-based

Screen structure & content

Views. My Tasks / Team / Project.

Fields. Title, due, priority, assignee, related lead/project, status.

User interactions

- 01** Create/edit/complete.
- 02** Filter.
- 03** Open related record.

Business logic

- Task can link to lead or project.
- Completed tasks retained.
- Overdue calculated, not manually set.

Technical implementation

- Task entity + indexes on assignee/due/status.

States

State	Required behavior
Loading	List skeleton.
Empty	No tasks message.
Error	Retry.
Success	Completion time recorded.

Responsive & motion

- Mobile checklist/cards.

Acceptance criteria

- Overdue logic correct.
- Assignee permission tested.

Quotations & Invoices

Optional V2 finance workflow for commercial follow-through.

Area	Specification
Primary user	Admin/Finance/Sales
Primary goal	Track commercial documents
Primary CTA	[NEW QUOTE]
Secondary CTA	[NEW INVOICE]
Data owner	Finance module
Permission	Finance/Admin

Screen structure & content

Quotes. Draft, Sent, Accepted, Rejected/Expired.

Invoices. Draft, Sent, Part Paid, Paid, Overdue, Cancelled.

Payments. Reference, amount, date, method.

User interactions

- 01** Create quote.
- 02** Generate PDF.
- 03** Send manually/integration.
- 04** Convert accepted quote to invoice.
- 05** Record payment.

Business logic

- Amounts/taxes require final business/tax rules.
- Do not enable automated tax assumptions without confirmation.
- Invoice numbers unique and immutable once issued.

Technical implementation

- Quote/Invoice/LineItem/Payment entities.
- PDF service.
- Optional email adapter.

States

State	Required behavior
Loading	List skeleton.

Empty

Create-first-document action.

Error

Draft remains saved.

Success

Document version/status recorded.

Responsive & motion

- Mobile document list/detail.

Acceptance criteria

- Numbering unique.
- Totals tested.
- Permissions strict.

Admin CMS & Media Studio

The missing core: non-technical content operations for videos, photos, thumbnails, colour-grade assets, reels and new portfolio projects.

CMS Overview

Give content owner one place to see drafts, recently uploaded media, processing jobs and publish tasks.

Area	Specification
Primary user	Owner / Content Editor
Primary goal	Publish fresh work safely
Primary CTA	[NEW PROJECT]
Secondary CTA	[UPLOAD MEDIA]
Data owner	CMS
Permission	Content/Admin

Screen structure & content

Quick actions. + New Project / + Upload Video / + Upload Photos / + New Reel / + Manage Thumbnails.

Processing. Uploads transcoding, failed jobs, items needing posters.

Drafts. Unpublished projects/reels.

Recently published. Latest live content.

User interactions

- 01** Start upload wizard.
- 02** Open failed processing job.
- 03** Open draft.
- 04** Preview public content.

Business logic

- Processing failures never auto-publish.
- Public visibility requires explicit publish.
- Drafts autosave where practical.

Technical implementation

- Content dashboard aggregates MediaAsset and PortfolioProject statuses.
- Processing job status from queue/worker.

States

State	Required behavior
Loading	Module skeleton.

Empty	Quick-start guidance.
Error	Independent widget errors.
Success	Every item links to actionable detail.

Responsive & motion

- Responsive cards/list.
- Wave can visualize processing queue progress subtly.

Acceptance criteria

- Owner can find failed media.
- No hidden publish state.

Media Library

Organize all reusable website/public media and internal project media without exposing storage complexity.

Area	Specification
Primary user	Content/Production
Primary goal	Find and reuse media
Primary CTA	[UPLOAD MEDIA]
Secondary CTA	[FILTER]
Data owner	Media subsystem
Permission	Content/Production/Admin

Screen structure & content

Filters. Video / Photo / Thumbnail / Poster / Raw Preview / Final Grade / Project / Visibility / Processing status.

Grid/list. Preview, filename/title, duration/dimensions, project, usage, status.

Actions. Open, replace poster, edit metadata, attach to project, archive/delete when safe.

User interactions

- 01 Search/filter.
- 02 Preview asset.
- 03 Edit metadata.
- 04 Attach/detach.
- 05 Archive.
- 06 Start upload.

Business logic

- Deletion blocked when asset is in active published use unless replaced/unlinked.
- Internal and Public visibility are distinct.
- Original storage key never shown to public API.

Technical implementation

- MediaAsset table + MediaUsage relation.
- Signed admin preview URLs.
- Thumbnail derivatives stored separately.

States

State	Required behavior
-------	-------------------

Loading	Aspect-ratio placeholders.
Empty	Upload first media.
Error	Retry filters.
Success	Usage count accurate.

Responsive & motion

- Mobile uses compact list/grid toggle.
- Long metadata hidden behind detail drawer.

Acceptance criteria

- Published-use protection works.
- Search metadata accurate.

New Video Upload Wizard

Let a non-technical owner upload a new video, generate web-ready variants, assign metadata, create/select thumbnails and attach it to a project or reel.

Area	Specification
Primary user	Owner / Content Editor
Primary goal	Add video without developer
Primary CTA	[UPLOAD & PROCESS]
Secondary CTA	[SAVE DRAFT]
Data owner	Media CMS
Permission	Content/Admin

Screen structure & content

Step 1 · Source. Choose file [MP4/MOV/etc according to final policy]. Show recommended maximum source size [TBD].

Step 2 · Purpose. Portfolio film / Reel / Showreel / BTS / Before-After Raw / Before-After Final / Internal Project Media.

Step 3 · Metadata. Title, project, category, description/caption, shoot date, client [optional], visibility.

Step 4 · Upload. Progress, speed, resumable state if supported.

Step 5 · Processing. Validate, probe codec/duration, transcode web variants, generate frame candidates.

Step 6 · Thumbnail. Choose generated frame or upload custom poster; configure desktop/mobile crop/focal point.

Step 7 · Colour grade metadata. Asset role: Raw / Graded Final / Neutral / Other. Colour space/profile [optional metadata]. Pair with counterpart if before/after.

Step 8 · Preview. Desktop/mobile player preview, poster, title, project association.

Step 9 · Publish decision. Save to Library / Add to Draft Project / Publish through project/reel workflow.

User interactions

- 01 Select/upload source.
- 02 Pause/resume if supported.
- 03 Retry failed chunk/job.
- 04 Select poster frame.
- 05 Upload custom thumbnail.
- 06 Pair raw/final.
- 07 Preview.
- 08 Save.

Business logic

- Upload is not considered ready until processing status=Ready.

- Do not publish source master directly.
- Required poster for public autoplay/portfolio video.
- Pairing Raw ↔ Final is optional and explicit.
- Colour-grade metadata never alters the master in MVP.
- Failed processing keeps source/job for retry according to retention policy.
- Public video requires rights/approval confirmation checkbox [TBD business process].

Technical implementation

- Use direct signed multipart/resumable upload to object/video storage.
- Backend creates MediaAsset with status=Uploading before issuing upload session.
- On completion, enqueue MediaProcessJob.
- Worker probes media using ffprobe or provider metadata, creates optimized renditions/poster candidates.
- Webhook/job callback updates MediaAsset=Ready/Failed.
- Poster selection creates PosterAsset/derivative reference.
- Public API exposes CDN playback URLs, never storage credentials.

States

State	Required behavior
Loading	Upload progress with bytes/percent; processing has separate progress/status.
Empty	No file selected; clear supported-format guidance.
Error	Validation, upload, processing and poster errors are distinct; retry without re-entering metadata.
Success	Ready asset appears in Media Library and can be attached immediately.

Responsive & motion

- Wizard step progress can use a thin waveform line; completed segments settle into accent color.
- Mobile admin supports upload but heavy source files may be discouraged with guidance.
- Reduced motion uses static progress.

Acceptance criteria

- Owner uploads without code/deployment.
- Invalid type/size rejected server-side.
- Interrupted upload recoverable if selected technology supports it.
- Public rendition plays on target browsers.
- Poster required before public publish.
- Raw/final pairing persists.

Batch Photo Upload

Allow fast upload and organization of project photography/BTS while preserving crops, metadata and accessibility.

Area	Specification
Primary user	Content Editor
Primary goal	Publish photos efficiently
Primary CTA	[UPLOAD PHOTOS]
Secondary CTA	[SAVE TO LIBRARY]
Data owner	Media CMS
Permission	Content/Admin

Screen structure & content

Step 1. Select/drop multiple images.

Step 2. Batch defaults: project, category, visibility, shoot date, photographer [optional].

Step 3. Per-image review: title, alt text, caption, focal point, orientation.

Step 4. Derivatives: thumbnail, web large, web medium; original retained privately according to policy.

Step 5. Attach to gallery/project or save to library.

User interactions

- 01** Add/remove files.
- 02** Apply batch metadata.
- 03** Edit per-image alt/crop.
- 04** Reorder gallery.
- 05** Save.

Business logic

- Public image needs meaningful alt text unless explicitly decorative.
- Large originals not loaded directly on public pages.
- Duplicate filename is allowed; storage uses unique key.

Technical implementation

- Direct upload + image processing worker/provider.
- Generate responsive derivatives.
- Store width/height/focal point/alt.

States

State	Required behavior
Loading	Per-file progress.
Empty	Drop-zone guidance.
Error	One failed file does not cancel all successful uploads.
Success	Batch summary lists successful/failed items.

Responsive & motion

- Mobile supports camera/file picker but desktop recommended for bulk.

Acceptance criteria

- Partial failure handled.
- Derivatives generated.
- Alt/focal data stored.

Thumbnail & Poster Manager

Control how every public video/project looks before playback across desktop, mobile and social previews.

Area	Specification
Primary user	Content Editor
Primary goal	Guarantee high-quality first impression
Primary CTA	[SAVE THUMBNAIL]
Secondary CTA	[GENERATE NEW FRAMES]
Data owner	Media CMS
Permission	Content/Admin

Screen structure & content

Source. Generated frame strip from video + Upload Custom.

Desktop crop. 16:9 / project layout preview.

Mobile crop. 4:5 / 9:16 depending component.

Focal point. Drag focal point; safe-area overlay.

Text overlay. Prefer none in asset; UI text belongs in HTML. Optional exported social thumbnail is separate later.

Preview. Home card, Work card, Project hero, Reel tile.

User interactions

- 01 Scrub frame candidates.
- 02 Choose frame.
- 03 Upload custom.
- 04 Set focal point.
- 05 Preview placements.
- 06 Save.

Business logic

- Each public video has one primary poster; layout-specific crop metadata may differ.
- Changing poster updates references without changing video.
- Published pages should revalidate/cache-bust after poster update.

Technical implementation

- Frame extraction via processing worker/provider.
- Poster derivative records with crop/focal metadata.
- Trigger frontend revalidation on save.

States

State	Required behavior
Loading	Preview placeholders.
Empty	Require choose/upload before public publish.
Error	Keep previous poster active until new save succeeds.
Success	All placement previews update.

Responsive & motion

- Touch-friendly focal-point editor.
- No hover-only controls.

Acceptance criteria

- Poster update visible after cache refresh.
- Previous live poster not broken on failed save.

Colour Grade / Before-After Manager

Manage raw-versus-final storytelling assets and colour metadata without pretending the CMS replaces DaVinci Resolve or professional grading.

Area	Specification
Primary user	Content/Editor
Primary goal	Showcase grading craft
Primary CTA	[SAVE COMPARISON]
Secondary CTA	[PREVIEW]
Data owner	Media CMS
Permission	Content/Admin

Screen structure & content

Pairing. Select Raw asset + Final graded asset.

Metadata. Colour space [e.g. Rec.709], camera/log profile [optional], grade note, colourist [optional].

Preview. Side-by-side, wipe slider, synced video playback if both are videos.

Public use. Select where this comparison may appear: Editing page / Project case study / hidden.

Optional future. Apply non-destructive LUT preview to a proxy only - not production master.

User interactions

- 01 Select pair.
- 02 Sync preview.
- 03 Set default comparison position.
- 04 Save public placement.

Business logic

- Raw and Final remain separate immutable media records.
- Pair must share compatible media type and preferably matching duration/dimensions for video comparison.
- Public comparison never exposes camera master; use proxy/raw-preview derivative.
- Optional LUT preview is a future worker feature, explicitly marked Preview.

Technical implementation

- BeforeAfterPair entity references rawMediaId/finalMediaId.
- Validation checks type, duration tolerance and dimensions.
- Synchronized player uses normalized timeline.

States

State	Required behavior
Loading	Both posters shown as available.
Empty	Prompt to select two assets.
Error	Mismatch warning; save blocked only for incompatible pair.
Success	Comparison available to approved CMS blocks.

Responsive & motion

- Mobile uses wipe/toggle; desktop can use synced split view.

Acceptance criteria

- No destructive processing.
- Raw master protected.
- Pair sync acceptable.

New Portfolio Project Wizard

Create a complete public case study from project information and previously uploaded or newly uploaded media.

Area	Specification
Primary user	Content Editor/Owner
Primary goal	Publish polished case study
Primary CTA	[PUBLISH PROJECT]
Secondary CTA	[PREVIEW]
Data owner	Portfolio CMS
Permission	Content/Admin

Screen structure & content

Step 1 · Basics. Title, slug, client, category, date/year, location, services, status Draft.

Step 2 · Story. Short summary, full narrative, creative challenge, outcome.

Step 3 · Cover. Select/upload hero video/image + required poster + mobile crop.

Step 4 · Content blocks. Add Video / Gallery / Text / BTS / Before-After / Testimonial / Credits.

Step 5 · Credits. Director, cinematography, editor, colour, drone, music/licensing [as applicable].

Step 6 · Related. Related service, next project preference, featured toggle/order.

Step 7 · SEO. SEO title, description, OG image, canonical defaults.

Step 8 · Preview. Desktop/tablet/mobile preview using real templates.

Step 9 · Publish. Validation summary, rights confirmation, Publish Now / Schedule [optional] / Save Draft.

User interactions

- 01** Autosave draft.
- 02** Add/reorder blocks.
- 03** Attach existing media.
- 04** Upload new media inline.
- 05** Preview.
- 06** Publish/unpublish/archive.

Business logic

- Required before publish: title, slug, category, cover, poster, at least one content block, approved status/rights checkbox.
- Featured projects may require home-safe cover ratio.
- Publishing never changes internal operational Project stage.
- One operational CRM project may link to zero/one public portfolio project.

Technical implementation

- PortfolioProject separate from OperationalProject or shared carefully with public-content layer; recommended separate content entity linked by optional OperationalProjectId.
- Structured ProjectBlock table/JSON with controlled block types.
- Publish transaction changes status and triggers frontend revalidation.
- Slug uniqueness enforced.

States

State	Required behavior
Loading	Autosave status visible.
Empty	Wizard starts with basics.
Error	Failed autosave shows persistent warning and keeps local edits.
Success	Published URL and View Live action.

Responsive & motion

- Mobile supports edits but full composition preview optimized for desktop.
- Block reorder touch-friendly.

Acceptance criteria

- Publish validation exact.
- Slug unique.
- Unpublish removes from public API.
- Preview never publicly indexed.

New Reel / Short-form Upload

Publish short-form work quickly while keeping video, poster and metadata consistent.

Area	Specification
Primary user	Content Editor
Primary goal	Add new reel quickly
Primary CTA	[PUBLISH REEL]
Secondary CTA	[SAVE DRAFT]
Data owner	Reels CMS
Permission	Content/Admin

Screen structure & content

Video. Upload or select Media Library video.

Poster. Select generated/custom thumbnail.

Metadata. Title, caption, category, platform reference URL, project association.

Display. Featured, order, publish status.

Preview. Website reel tile + mobile crop.

User interactions

01 Upload/select.

02 Set poster.

03 Preview.

04 Publish.

Business logic

- Reel publication requires Ready video + poster.
- External Instagram link optional; website does not depend on social embed.

Technical implementation

- Reel entity references MediaAsset.
- Frontend uses optimized playback URL.

States

State	Required behavior
Loading	Processing status.
Empty	Select/upload guidance.

Error

Draft retained.

Success

Reel appears in configured public section.

Responsive & motion

- 9:16 preview emphasized; desktop placement may crop carefully.

Acceptance criteria

- Reel works without Instagram.
- Poster/crop valid.

Showreel Manager

Let owner update the primary showreel without developer deployment while protecting hero performance.

Area	Specification
Primary user	Owner/Content
Primary goal	Refresh showreel safely
Primary CTA	[SET ACTIVE]
Secondary CTA	[PREVIEW]
Data owner	Site CMS
Permission	Content/Admin

Screen structure & content

Current. Active showreel + poster + duration + last updated.

Replace. Select existing Ready video or upload new.

Poster. Required.

Hero usage. Use as hero background? Yes/No; if yes, require lighter optimized hero variant.

Fallback. Static poster/fallback project asset.

User interactions

- 01 Preview.
- 02 Set active.
- 03 Rollback to previous active.

Business logic

- Only one active showreel.
- Activation triggers revalidation.
- Previous media remains in library for rollback.

Technical implementation

- SiteSetting or ShowreelVersion entity.
- Atomic active switch.

States

State	Required behavior
Loading	Preview poster.

Empty	Fallback hero configured.
Error	Previous active remains.
Success	New showreel live after cache refresh.

Responsive & motion

- Hero background may use separate low-bitrate rendition.

Acceptance criteria

- Atomic switch verified.
- Rollback works.

End-to-End Project Workflow

Connect sales, production, editing and publication without mixing internal and public states.

G1. Operational lifecycle

Stage	Business actions	System outputs
Lead Won	Confirm client/scope	Client + Project created
Pre-production	Brief, schedule, crew, shot list	Tasks/events/files
Scheduled	Lock dates/location	Calendar events
Production	Shoot completed	Milestone + internal media references
Editing	Edit/grade/sound	Tasks, review files
Client Review	Collect feedback	Notes/revisions
Final Delivery	Deliver masters	Delivery milestone
Completed	Close project	Eligible for portfolio consideration
Portfolio	Create public case study separately	PortfolioProject Published

G2. Key separation

Operational Project ≠ Public Portfolio Project. A completed client job may never be public. Publishing is a separate deliberate content decision.

G3. Conversion transaction

- 01** Authorized user opens Won lead.
- 02** System checks whether client already exists by linked client or matching confirmed contact.
- 03** Create/reuse Client.
- 04** Create OperationalProject with copied lead scope context.
- 05** Link Lead.ConvertedProjectId and set conversion timestamp.
- 06** Create default milestones/tasks from project template [optional].
- 07** Write ActivityLog.
- 08** Return Project URL.

Technical Architecture

A scalable but lean architecture for the first production release.

H1. Recommended topology

Next.js Public + Admin → ASP.NET Core Web API → PostgreSQL/SQL Server → Object/Video Storage + CDN → Processing Worker/Provider → Email/Notification Adapter

Layer	Recommended responsibility
Next.js	Public SSR/SEO, admin UI, motion, forms, previews
ASP.NET Core	Auth, authorization, CMS/CRM business logic, upload sessions, leads, publishing
Database	Structured data, states, relationships, audit, metadata
Object/video storage	Source/proxy/renditions/posters; no video blobs in DB
CDN	Fast global public delivery
Worker / media provider	Transcode, frame extraction, image derivatives, metadata probing
Queue	Media processing, notification jobs, revalidation events
Email provider	Transactional lead/admin messages

H2. Suggested deployment split

- Public/admin Next.js can share one app/repo or be separate route groups.
- ASP.NET Core API deploys independently.
- Database managed when budget permits.
- Storage/CDN externalized from API server.
- Staging and production use separate databases/storage buckets/prefixes.

H3. Upload architecture

- 01** Admin asks API to create upload session.
- 02** API validates permission/purpose and creates MediaAsset=PendingUpload.
- 03** API returns signed direct-upload credentials/session.
- 04** Browser uploads directly to storage/provider, not through API memory.
- 05** Completion callback/API marks UploadComplete and queues processing.
- 06** Worker/provider creates renditions/posters/metadata.
- 07** Callback updates MediaAsset=Ready or Failed.
- 08** Admin selects poster/crop/pairing and attaches asset.
- 09** Publish validation checks Ready state.

Data Model & API Contracts

Developers should implement stable entities and explicit business transitions.

I1. Core entities

Entity	Purpose	Critical fields
User	Internal identity	Id, Email, Role, Status, MFA/Provider
Lead	Sales opportunity	Service, brief, contact, source, status, assignee
Client	Customer	Name, contacts, notes, archive status
OperationalProject	Delivery work	Client, stage, dates, service, owner
Task	Work item	Assignee, due, status, related entity
CalendarEvent	Schedule	Type, start/end, timezone, related entity
PortfolioProject	Public case study	Slug, story, cover, blocks, status, SEO
ProjectBlock	Controlled content block	Type, order, content/media refs
MediaAsset	Media metadata	Kind, role, status, visibility, storage refs, dimensions/duration
MediaDerivative	Renditions/posters	AssetId, kind, URL/key, dimensions/bitrate
BeforeAfterPair	Raw/final comparison	RawMediaId, FinalMediaId, metadata, public placement
Reel	Short-form public content	MediaId, poster, caption, status, order
Service	Offer	Name, description, active, sort order
Package	Price anchor	Service, display price, inclusions, active
Testimonial	Proof	Quote, person, approval/status
ActivityLog	Audit	Actor, action, entity, timestamp, metadata

I2. API surface - MVP

Method	Route	Business action
GET	/api/public/projects	Published portfolio
GET	/api/public/projects/{slug}	Case study
GET	/api/public/services	Services/packages
POST	/api/public/leads	Create lead
POST	/api/auth/login	Admin login
GET	/api/admin/leads	Lead list/filter
PATCH	/api/admin/leads/{id}/status	Progress lead

POST	/api/admin/leads/{id}/convert	Lead→Client+Project
GET/POST	/api/admin/projects	Operational projects
GET/POST	/api/admin/portfolio	Public projects
POST	/api/admin/media/upload-session	Create signed upload
POST	/api/admin/media/{id}/complete	Confirm upload / enqueue
PATCH	/api/admin/media/{id}	Metadata/poster/visibility
POST	/api/admin/before-after	Create comparison
POST	/api/admin/portfolio/{id}/publish	Publish/revalidate

I3. MediaAsset state machine

PendingUpload → Uploading → Uploaded → Processing → Ready | Failed → Archived

- Only Ready can be selected for public playback.
- Failed can retry processing without re-upload when source retained.
- Archived is hidden from normal selection but preserved according to retention.

Security, Permissions & Data Protection

The public site is artistic; the admin is a business system and must be treated accordingly.

J1. Security baseline

- HTTPS everywhere.
- Secure session/cookie or token strategy with server-side authorization.
- Role-based policies on every admin endpoint.
- MFA for owner/admin when feasible.
- Rate limits on auth and public lead endpoints.
- Bot protection on lead form.
- Upload MIME/extension/size validation; never trust filename.
- Signed upload/download URLs; private source objects not world-readable.
- Secrets in environment/secret manager only.
- Structured logs without dumping sensitive form payloads unnecessarily.
- Automated DB backup + restore test.

J2. Permission matrix

Module	Admin	Content	Sales	Production	Finance
Site/CMS	Full	Edit/Publish	Read	Read	-
Media	Full	Upload/Edit	Read	Upload/Internal	-
Leads	Full	Read	Full	Read	Read limited
Clients	Full	Read	Full	Read	Read
Projects	Full	Read	Read	Full	Read
Finance	Full	-	Quote limited	Read limited	Full
Users/Settings	Full	-	-	-	-

SEO, Performance, Accessibility & Analytics

Premium visuals only work if the site loads, ranks and remains usable.

K1. Performance rules

- Never preload all portfolio videos.
- Hero uses poster immediately and an optimized video variant.
- Images use responsive derivatives and lazy loading below fold.
- Offscreen videos pause/unmount based on component behavior.
- Admin uploads go direct to storage, not through the API process memory.
- Code-split heavy admin/media tools and animation code.

K2. SEO

- Unique title/description per project and service.
- Sitemap from published routes.
- Canonical URLs and clean slugs.
- Open Graph images from selected poster/cover.
- VideoObject structured data where appropriate.
- LocalBusiness/Organization only with accurate real information.

K3. Conversion analytics

Event	Meaning
hero_start_project	Primary hero CTA
project_open	Portfolio engagement
showreel_play	Intentional play
editing_compare	Before/after interaction
lead_start	Lead form started
lead_step	Step completed
lead_success	Lead persisted
whatsapp_click	Direct contact click
package_quote	Package-based enquiry

Deployment, Hosting & Cost Model

Take the system from local development to a dependable international production deployment.

L1. Environments

Environment	Purpose	Rules
Local	Development	Seed/demo data, local media mocks or dev bucket
Staging	Client approval and owner training	Separate DB/storage, no real production leads unless agreed
Production	Live business	Backups, monitoring, controlled deployments

L2. Cost planning placeholders

Cost center	Initial planning placeholder	Scaling driver
Domain	₹[800-1,500]/year	TLD/registrar
Frontend hosting	₹[0-2,000+]/month	Commercial plan/traffic
API compute	₹[500-1,500+]/month	CPU/RAM/requests
Database	₹[0-2,000+]/month	Storage/connections/backups
Media storage/CDN	₹[USAGE-BASED]	GB stored + delivery
Transcoding	₹[USAGE-BASED]	Minutes/resolutions
Email/notifications	₹[USAGE-BASED]	Volume
Monitoring/backups	₹[TBD]	Retention/features

L3. Domain/DNS/SSL

- [DOMAIN] points via DNS to frontend hosting/CDN.
- API can use api.[DOMAIN] or same-domain reverse proxy.
- Admin can use /admin or admin.[DOMAIN].
- HTTPS certificate automated by hosting/reverse proxy.
- Do not run permanent production from a personal laptop.

Development Phases, QA & Handover

A step-by-step delivery plan that combines business validation with implementation gates.

M1. Phase-by-phase plan

Phase	Business work	Technical work	Exit criteria
01 Discovery	Confirm services, audience, lead questions, content rights, pricing policy	Repo/architecture decision, environments plan	Signed scope + placeholder list
02 UX	Sitemap, user flows, CRM workflows, CMS upload journeys	Route map, entity draft	Wireframes approved
03 Visual	Brand tokens, public art direction, CRM wave language	Design tokens/components/motion primitives	Key screens approved
04 Public Web	Real/sample content review	Next.js pages, SEO foundation, motion	Staging public site complete
05 Lead Engine	Qualification and follow-up rules	Lead API, validation, anti-bot, notifications	Lead appears in CRM reliably
06 CRM	Sales/production process validation	Dashboard/leads/clients/projects/tasks/calendar	Workflow acceptance
07 Media CMS	Owner upload process validation	Video/photo upload, processing, thumbnails, grade pairing	Owner uploads without dev
08 Portfolio CMS	Case-study content process	Project wizard, blocks, preview/publish	Owner publishes project
09 Integrations	Email/finance decisions	Providers/adapters	Integration tests
10 QA	Owner/team UAT	Security, performance, browser, responsive, a11y	Launch sign-off
11 Production	Account/billing ownership	DNS/SSL/deploy/backups/monitoring	Go live
12 Handover	Training, SOP	Docs, restore guide, maintenance	Owner self-sufficient

M2. Detailed QA checklist

- Public: hero/poster fallback, project filters, case-study blocks, CTA source context.
- Lead: validation, duplicate submit, persistence-before-notification, spam/rate limits.
- CRM: status transitions, assignment, notes/history, lead conversion idempotency.
- Project: stage/history, task/calendar linking, internal/public media separation.
- Video upload: type/size validation, interrupted upload behavior, processing success/failure, playback.
- Photo upload: partial failure, derivatives, alt text/focal point.
- Thumbnail: frame extraction, custom poster, mobile/desktop crop, failed-save rollback.
- Colour-grade pair: raw/final compatibility, sync preview, public visibility.
- Portfolio project: autosave, block reorder, preview, publish validation, unpublish.
- Security: authorization, source media privacy, signed URLs, secrets, backups.
- Responsive: 390 / 768 / 1024 / 1440 plus real mobile testing.
- Accessibility: keyboard, focus, labels, reduced motion, player controls.

M3. Owner handover SOP

- 01 Login and MFA/session basics.
- 02 Upload a new video from start to Ready.
- 03 Select/customize thumbnail and crops.
- 04 Upload a batch of photos and fix alt/focal data.
- 05 Create a Raw/Final colour-grade comparison.
- 06 Create a new portfolio project and attach media.
- 07 Preview desktop/mobile and publish.
- 08 Create/publish a reel.
- 09 Review a new lead, add note and progress status.
- 10 Convert Won lead into Client + Operational Project.
- 11 Update project milestones/tasks/calendar.
- 12 Understand backups, support and what requires a developer.

M4. Final definition of done

The product is complete when a prospect can discover the studio, submit a qualified brief, the team can progress that lead into a project, and the owner can independently upload new videos/photos/thumbnails/colour-grade pairs and publish new work without touching code or deployment.